EviMatch: Evidence-Grounded and Bias-Audited Resume–Job Matching, with a Leakage Analysis of a Public Benchmark

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Abstract—Automated resume screening is widespread and increasingly regulated, yet published matchers usually return an opaque score, depend on private application logs, and leave bias unmeasured. We present EviMatch, an open and deterministic pipeline that removes identity cues before scoring, parses a posting into weighted requirements, links each requirement to the resume sentence that best supports it, and combines lexical, semantic, skill and experience features in a low-capacity head. On a public benchmark of 8,000 labelled pairs we first show that the official split is leaky: 99.9% of test resumes also occur in training, and a predictor that ignores the posting reaches macro-F1 0.42. Under a doubly-disjoint protocol, in which neither resume nor posting of a test pair is seen in training, EviMatch reaches macro-F1 0.473 against 0.405–0.412 for TF-IDF, BM25, dense-similarity and supervised embedding baselines (paired differences 0.06–0.07, intervals excluding zero). Ablations indicate that experience and skill features carry the signal, whereas semantic coverage supplies evidence rather than accuracy. Deleting the sentences an explanation cites changes the score about four times more than deleting random text. A counterfactual audit found and led us to fix a defect in our own redactor; afterwards, name cues change the score by exactly zero, while proxy cues such as affinity organisations still shift it by up to 0.2 points on a 100-point scale. Absolute accuracy is modest and labels are noisy, so the system supports review and should not automate decisions.

Index Terms—resume screening, person–job fit, explainable AI, algorithmic fairness, sentence embeddings, benchmark leakage

I. Introduction

Employers routinely receive far more applications than people can read, and software now decides which resumes reach a human. Screening software is therefore a consequential system, and regulators have started to treat it as one: the European AI Act places systems that filter or evaluate job applications in its high-risk category, with obligations on transparency and human oversight [1]. Two properties follow for any practical matcher. A reviewer must be able to see why a candidate was scored as they were, and the builder must be able to show that irrelevant attributes such as a name do not change the outcome.

Research on person–job fit has mostly pursued accuracy. Convolutional and recurrent matchers learn from historical application logs [2], [3]; fine-tuned sentence encoders use hundreds of thousands of consultant-labelled pairs [4]; and recent work adds language-model data augmentation [5], [6] or multi-agent prompting [7]. These systems share three weaknesses for deployment. Their training data are private, so results cannot be reproduced. Their output is usually one similarity score, with no statement of which requirement is met by which part of the resume. And bias is treated as a future direction or left unmeasured, although audits show that embedding retrievers and language models rank resumes differently when only the name changes [8]–[11]. Removing explicit gender or race indicators is a common response, yet it is known to leave proxy signal behind [12], [13].

A further, less discussed problem concerns evaluation. The public resume–job fit benchmark we use contains 8,000 labelled pairs but only 643 distinct resumes and 351 distinct job descriptions. Any single random or official split therefore lets a model profit from having seen the same resume before. A scorer that never looks at the job description can already be competitive on such a split.

This paper makes four contributions.

- EviMatch, an open, deterministic matching pipeline that (i) removes identity cues before scoring, (ii) parses a posting into weighted requirements, (iii) links each requirement to its best-supporting resume sentence, and (iv) combines lexical, semantic, skill and relational experience features in a low-capacity head whose every signal can be inspected.
- A leakage analysis of the benchmark and a doubly-disjoint evaluation protocol in which neither the resume nor the job description of a test pair occurs in training.
- Empirical findings with confidence intervals: which feature groups carry the signal, how little labelled data is needed, and that cited evidence is faithful to the model under a deletion test.
- A counterfactual bias audit for explicit and proxy cues. The audit exposed a defect in our own redactor, which we report and fix; the corrected audit quantifies what redaction does and does not remove.

Section II reviews related work, Section III describes the system, Section IV the experimental protocol, Section V the results, and Section VI concludes with limitations and concrete next steps.

II. Related Work

A. Person–job fit and resume matching

Early neural matchers treat fit as supervised learning on application histories. PJFNN encodes job requirements and candidate experience with convolutional networks and can indicate which requirement items a candidate satisfies [2]; APJFNN adds hierarchical attention over abilities [3]. Both rely on large, proprietary logs. Sentence encoders removed some of that burden: conSultantBERT fine-tunes a Siamese SBERT on more than 270,000 consultant-labelled pairs [4], and an end-to-end BERT pipeline ranks candidates after parsing resumes [14]. ConFit and ConFit v2 address label scarcity with LLM-generated augmentations and contrastive learning, improving ranking over BM25 and commercial embeddings [5], [6]. On the systems side, Smart-Hiring combines entity extraction with embeddings in an inspectable pipeline [15], and a multi-agent LLM framework scores resumes against HR ratings [7]. Table I summarises the properties that matter for the setting studied here.

B. Retrieval building blocks

Sentence-BERT made dense similarity practical [16], and distilled encoders such as MiniLM [17] and general-purpose embedding families [18] give small, fast models. On heterogeneous retrieval tasks, BM25 [19] remains a strong zero-shot baseline [20], which is why we report lexical baselines alongside dense ones. ColBERT's late interaction scores a query by matching each query unit to its best document unit [21]; our requirement-level coverage applies the same idea at sentence level so that the best match can be displayed to a reviewer.

C. Skill extraction

SkillSpan annotates hard and soft skill spans in English postings and is the reference benchmark for learned extractors [22]. Such extractors need training data or model access. We instead use a small transparent lexicon so that every extracted skill can be audited, and we state the coverage trade-off in Section VI.

D. Bias in automated screening

The audit tradition goes back to a field experiment in which identical resumes with different names received different callback rates [23]. Applying the same design to embedding-based retrieval, Wilson and Caliskan found strong preference for white-associated names [8]; audits of GPT and other language models report name- and gender-dependent scores or decisions [9]–[11]. Scrubbing explicit indicators reduces but does not eliminate dependence, because proxies persist [12], [13], and reviews of vendor claims urge evidence rather than assertion [24]. We follow this audit design and report residual proxy effects instead of claiming invariance.

III. The EviMatch System

Figure 1 gives the overview. Given a resume R and a job description J, the system returns a fit score $S \in [0, 100]$, a decision band, and an evidence record.

A. Blind-screening front end

Redaction is rule based, deterministic and runs before segmentation. It removes e-mail addresses, telephone numbers, URLs, header names and "Name:" fields, personal-detail fields (date of birth, marital status, nationality, and similar), honorifics, graduation dates, institution names, and organisation names that carry a gender marker. Gendered pronouns and role nouns are neutralised ("she" $\rightarrow$ "they"). Institution names are removed because institutional prestige is a proxy for socio-economic background; for the same reason the system uses neither college tier nor grade point average, both of which an earlier version of this project fed to a language model. A redaction report lists how many items of each class were removed, never their content.

B. Requirement parsing

A posting mixes duties and qualifications with benefits, legal text and company marketing, which dilutes any whole-document similarity. The parser splits J into sentence units, assigns each a kind $\kappa \in \{\text{required}, \text{preferred}, \text{duty}, \text{other}\}$ from section headers and cue phrases, and drops boilerplate. Weights are fixed constants, w_{required}=1, w_{duty}=0.75, $w_{\text{preferred}}$=0.5, w_{other}=0.4.

C. Requirement-level evidence matching

Let $q_1, \ldots, q_K$ be the requirement units with weights w_k, and $s_1, \ldots, s_M$ the sentence units of the redacted resume. With $\mathbf{e}(\cdot)$ an L2-normalised sentence embedding,

$$c_k = \max_m \ \mathbf{e}(q_k)^\top \mathbf{e}(s_m), \qquad \text{cov}(R, J) = \frac{\sum_k w_k c_k}{\sum_k w_k}. \quad (1)$$

The arg-max sentence is the evidence for requirement k. A requirement is shown as evidenced if $c_k \geq 0.50$, partial if $0.40 \leq c_k < 0.50$ and otherwise without evidence. The two cut-offs were fixed from training-split probes and not tuned afterwards. Because c_k is an embedding similarity it indicates topical relevance, not demonstrated competence, and the interface says so.

D. Features and fit model

Each pair yields 22 relational features in four groups: lexical (TF-IDF cosine, length-normalised BM25 with the posting as query, IDF-weighted term overlap); semantic (document cosine, the weighted coverage of Eq. (1), its required/preferred variants, a lower quartile and minimum over requirements, and the fractions of requirements above the two cut-offs); skills (IDF- and weight-adjusted coverage of the posting's skills by the resume, Jaccard overlap, and the number of shared and missing required skills,

TABLE I
Selected prior work and the property this paper adds. "Training data" is as stated by the cited paper; "–" means the paper's abstract does not state it.

Work	Method	Training data	Evidence shown	Gap relevant to this paper
PJFNN [2]	CNN, joint representation	Historical applications	Requirement items satisfied	Needs private logs; no bias audit stated
APJFNN [3]	RNN + ability-aware attention	Historical applications	Attention weights	Same data dependence; no bias audit stated
conSultantBERT [4]	Fine-tuned Siamese SBERT	>270k expert-labelled pairs	Single similarity	Large labelled corpus; no requirement-level evidence
Bhatia et al. [14]	BERT sentence-pair ranking	Parsed resumes	–	Job text approximated by past experience text
ConFit / v2 [5], [6]	LLM augmentation + contrastive	Interaction-labelled pairs	Single similarity	Needs an external LLM at training time
Smart-Hiring [15]	NER + embeddings	Real resumes and postings	Extracted entities	Bias mitigation listed as future work
Lo et al. [7]	Multi-agent LLM + retrieval	Online resumes (HR ratings)	LLM rationale	LLM cost and run-to-run variation; no demographic audit stated
EviMatch (this work)	Requirement-level evidence + low-capacity head	Public benchmark, 8k pairs	Per-requirement sentence, skills, drivers	Audit of explicit and proxy cues; leakage-aware evaluation

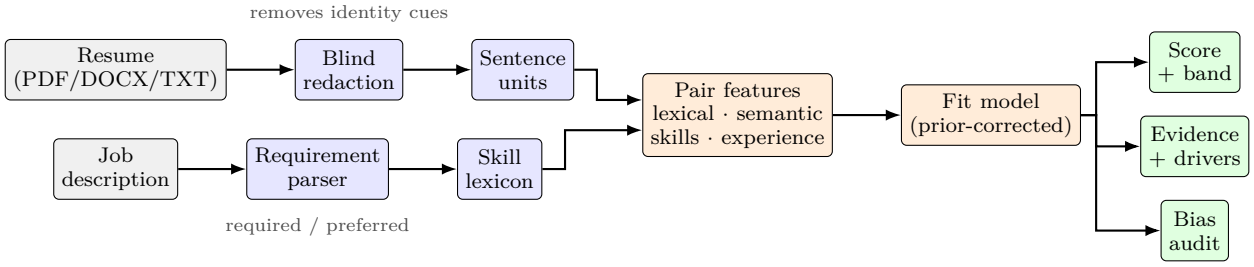


Fig. 1. EviMatch pipeline. Identity cues are removed before any feature is computed; the same analysis object yields the feature vector and the evidence shown to the reviewer, so an explanation cannot describe a signal the model did not use.

from a 350-entry transparent lexicon); and experience (estimated years of experience minus the years the posting asks for, an indicator that the requirement is met, and the same pair for education level). Features that describe only the resume or only the posting (for example years of experience on its own) are excluded; the next section explains why.

The head is a three-class model over {No, Potential, Good Fit}. We train it without class re-weighting, so its probabilities $P(c \mid x)$ stay calibrated to the base rate, and decide by dividing by the training prior π_c:

$$\hat{c} = \arg\max_c \; P(c \mid x)/\pi_c, \tag{2}$$

$$S = 100\,\big(P(\text{Good} \mid x) + \tfrac{1}{2}P(\text{Potential} \mid x)\big). \tag{3}$$

Eq. (2) is the Bayes rule for balanced accuracy and maps to the bands Reject, Review and Shortlist; the middle band is where human review is expected. Candidate heads are a multinomial logistic regression with an L_2 penalty and a histogram gradient-boosted classifier [25]; selection uses the one-standard-error rule described in Section IV.

E. Driver attribution

The contribution of feature f to a candidate's score is the occlusion effect

$$\Delta_f = S(x) - S(x_{f \leftarrow \mu_f}), \tag{4}$$

where μ_f is the training mean. Occlusion is model agnostic, deterministic and needs no extra library; effects are also summed per feature group for display. Section V checks it with a deletion test rather than assuming it is faithful.

F. Counterfactual bias audit

For a resume R the audit builds variants that change one irrelevant cue and compares the score with a control. Name cells: a name from each of ten race/ethnicity × gender cells (four names per cell) is prepended as a header. Proxy cues: a final line is appended, always compared with a neutral line of similar length: an affinity-organisation membership (three groups), stereotyped hobbies (two codings), a career break for childcare versus for study, and a gender-marked

organisation that the redactor is designed to remove. The statistics are the mean score change with a job-description-cluster bootstrap interval, the decision-flip rate, and the largest gap between name cells. Name-to-group association is statistical and imperfect; the lists are fixed in the code so the audit is reproducible.

IV. Experimental Setup

A. Data

We use the public Hugging Face benchmark cnamuangtoun/resume-job-description-fit: 6,241 training and 1,759 test pairs with labels No Fit (50%), Potential Fit (25%) and Good Fit (25%). The hub page publishes no dataset card or licence, and how the labels were produced is undocumented; we treat the benchmark as research data of uncertain provenance and verify file checksums before every run. The resumes in this data set are already anonymised in places (placeholder city and state), so the cost of redaction measured here is a lower bound for raw resumes.

B. Splits

S1 is the official split. It is job-description-disjoint, but 99.9% of test resumes also occur in training. S2 pools all 8,000 pairs, partitions resumes and job descriptions at random into five folds each (seed 13), and for fold k trains on pairs whose resume and job description fall outside fold k and tests on pairs whose resume and job description fall inside it. No test resume or posting is seen in training, and the five test sets are disjoint. Because both axes are held out, S2 has fewer candidates per posting than S1, so its ranking metrics are not comparable in level with S1.

C. Systems compared

B0 predicts the majority class. B0r predicts the training label majority of the resume and is a leakage diagnostic, not a competitor. B1–B4 score each pair with TF-IDF cosine, BM25, MiniLM [16], [17] document cosine or BGE-small document cosine [18] on raw text and learn only two score thresholds on training data. B6 trains logistic regression on the concatenation $[\mathbf{u} \odot \mathbf{v}, |\mathbf{u} - \mathbf{v}|, \cos]$ of whole-document MiniLM embeddings. M1 uses lexical, skill and experience features; M2 only semantic features; M3 is EviMatch with a linear head and M4 with a gradient-boosted head. X1/X2 add the excluded single-document features to M3/M4 to measure leakage.

D. Training and model selection

Hyper-parameters are chosen by five-fold grouped cross-validation on the training portion only, with folds grouped by job description. The grid is $C \in \{0.003, \ldots, 1\}$ for the linear head and depth $\{2, 3\}\times$ iterations $\{100, 200\}$ for the boosted head (learning rate 0.05). Under the one-standard-error rule we take the simplest candidate whose mean macro-F1 is within one standard error of the best. The sentence encoder is all-MiniLM-L6-v2 (22.7M parameters) with a 128-token limit per sentence unit. Experiments ran on an Apple M4 laptop (16 GB) in a few minutes per configuration; all seeds are 13 and library versions and the git commit are stored with every result file.

E. Metrics and uncertainty

The primary metric is macro-F1 over the three classes. We also report accuracy, one-vs-rest ROC-AUC for Good Fit, per-class precision/recall/F1, a confusion matrix, quadratic-weighted kappa, and for ranking nDCG@10 per posting (graded gains 0/1/2, postings with at least five candidates, seeded random tie-breaking). Intervals are 95% percentile bootstrap intervals that resample job descriptions, the unit of independence; differences between systems use paired resamples. The evaluation protocol, including every deviation made after it was written, is in the repository.

V. Results and Discussion

A. Accuracy on the official split and on the doubly-disjoint split

Table II and Fig. 2 give the main comparison. Three observations stand out.

The official split rewards memorising resumes. A predictor that ignores the job description and returns the training majority label of the resume (B0r) reaches macro-F1 0.420, above TF-IDF, BM25, and both dense similarity baselines (0.339–0.365), and EviMatch's linear head (M3, 0.439) is not significantly better than it (paired difference $+0.019$, 95% CI $[-0.044, +0.085]$). The supervised embedding head B6 reaches 0.480 on S1 but only 0.409 on S2, where resumes are held out. We therefore treat S2 as the primary measure of matching ability.

On S2, EviMatch is consistently better than similarity baselines. M3 reaches macro-F1 0.473 (95% CI 0.437–0.507) against 0.405–0.412 for B1–B4 and B6. Paired differences are $+0.061$ to $+0.068$ with intervals that exclude zero (all $p \leq 0.008$), and nDCG@10 is 0.838 against 0.782–0.805 (paired differences $+0.033$ to $+0.056$). The gradient-boosted head M4 is statistically indistinguishable from M3 (-0.006, $[-0.032, +0.023]$), so we keep the linear head, whose behaviour is easier to inspect.

Absolute accuracy is modest. On S2, M3 has Good-Fit precision 0.32 and recall 0.42, and a quadratic-weighted kappa of 0.30 (Table III, Fig. 3). The Potential-Fit class, the most ambiguous by construction, is confused with both neighbours. The model is useful for ordering candidates within a posting, not for automated rejection, and Section VI returns to this.

B. Where the signal comes from

Table IV and Fig. 4 remove one feature group at a time. On S2, removing the four experience features costs 0.053 macro-F1 (CI $[-0.076, -0.029]$) and removing skills costs 0.032 ($[-0.056, -0.008]$). Removing the lexical group

TABLE II

Main results. S1: official split. S2: doubly-disjoint five-fold evaluation. AUC_G: ROC-AUC for Good Fit vs. rest. nDCG@10 is the mean over postings with at least five candidates; S2 postings have fewer candidates, so ranking levels are not comparable across the two splits. B0r exploits resume overlap and is a diagnostic only. X1/X2 add single-document features that let a model memorise resumes.

	S1 official (1,759 pairs, 71 JDs)				S2 doubly disjoint (1,562 pairs, 318 JDs)			
System	macro-F1 [95% CI]	Acc.	AUC_G	nDCG@10	macro-F1 [95% CI]	Acc.	AUC_G	nDCG@10
B0 majority class	0.218 [0.201, 0.234]	0.487	0.500	0.532	0.226 [0.217, 0.236]	0.514	0.500	0.724
B0r resume-identity prior	0.420 [0.371, 0.466]	0.504	0.663	0.695	–	–	–	–
B1 TF-IDF cosine	0.365 [0.324, 0.407]	0.429	0.559	0.663	0.409 [0.381, 0.437]	0.463	0.627	0.805
B2 BM25	0.342 [0.299, 0.387]	0.393	0.473	0.643	0.405 [0.376, 0.433]	0.455	0.584	0.802
B3 MiniLM doc. cosine	0.339 [0.298, 0.386]	0.395	0.517	0.710	0.410 [0.377, 0.441]	0.456	0.603	0.790
B4 BGE doc. cosine	0.359 [0.318, 0.404]	0.402	0.495	0.715	0.412 [0.381, 0.444]	0.450	0.608	0.782
B6 embedding head (sup.)	0.480 [0.429, 0.525]	0.497	0.694	0.739	0.409 [0.376, 0.440]	0.444	0.597	0.791
M1 lexical+skills+exp.	0.438 [0.385, 0.489]	0.475	0.581	0.721	0.480 [0.443, 0.515]	0.507	0.641	0.831
M2 semantic only	0.346 [0.299, 0.390]	0.402	0.519	0.708	0.376 [0.345, 0.406]	0.430	0.597	0.788
M3 EviMatch (linear)	0.439 [0.384, 0.494]	0.482	0.562	0.744	0.473 [0.437, 0.507]	0.512	0.640	0.838
M4 EviMatch (GBM)	0.448 [0.396, 0.495]	0.492	0.605	0.754	0.467 [0.434, 0.498]	0.505	0.681	0.832
X1 M3 + identity feats.	0.464 [0.401, 0.524]	0.497	0.582	0.739	0.477 [0.439, 0.512]	0.516	0.667	0.818
X2 M4 + identity feats.	0.465 [0.408, 0.517]	0.511	0.633	0.772	0.480 [0.442, 0.514]	0.522	0.687	0.841

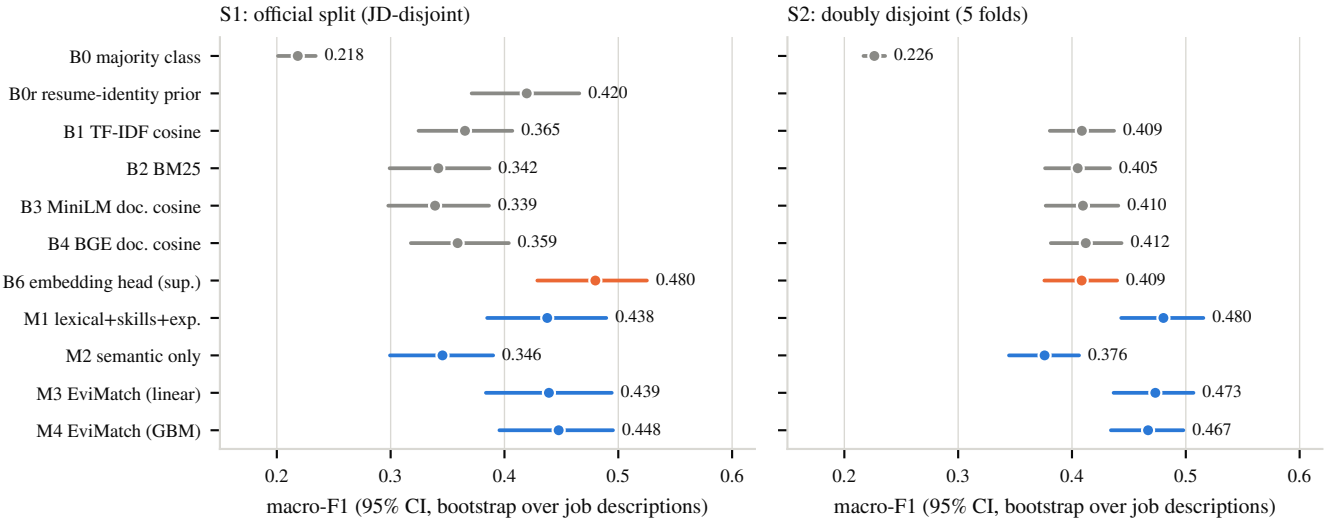


Fig. 2. Macro-F1 with 95% intervals from bootstrap resampling of job descriptions. Gray: similarity baselines and diagnostics; orange: supervised embedding head; blue: EviMatch variants.

TABLE III

Per-class results of M3 on S2.

Class	Precision	Recall	F1
No Fit	0.731	0.606	0.663
Potential Fit	0.380	0.404	0.391
Good Fit	0.322	0.422	0.365

changes nothing (0.000), and removing the semantic group is, if anything, slightly positive (+0.007, $[-0.015, +0.031]$). The four experience features alone reach 0.461, within noise of the full model (-0.012, $[-0.046, +0.026]$), although ranking quality then drops by 0.060 nDCG@10. In this benchmark the labels appear to track how well a candidate's seniority and listed skills line up with the posting, and the semantic coverage we expected to be central adds no significant accuracy. We still keep it, because it supplies the sentence-level evidence that makes the score reviewable, but we do not claim that it improves accuracy. Neither redaction (0.000 on S2, +0.002 on S1), the induced vocabulary, nor swapping MiniLM for BGE-small (+0.010, $[-0.010, +0.033]$) changed accuracy measurably. The redaction result is a lower bound on its cost because the benchmark's resumes are partly anonymised.

C. Leakage

Adding the seven single-document features to M3 (X1) raises S1 macro-F1 by 0.025 (0.439 to 0.464; paired -0.024, $[-0.056, +0.004]$) but changes S2 by only +0.004. The direction matches the explanation that these features let a model recognise resumes, but with 71 test postings the S1 increase is not statistically significant, so we report it

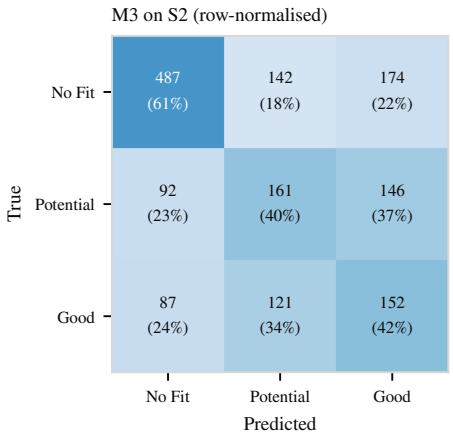


Fig. 3. Confusion matrix of M3 on S2 (counts and row percentages).

TABLE IV
Ablations of M3 (macro-F1 and difference from the full model).

	S1		S2	
Variant	macro-F1	Δ	macro-F1	Δ
full	0.439	+0.000	0.473	+0.000
minus lexical	0.434	-0.005	0.473	-0.000
minus semantic	0.438	-0.001	0.480	+0.007
minus skills	0.402	-0.037	0.441	-0.032
minus experience	0.394	-0.045	0.420	-0.053
only lexical	0.357	-0.082	0.384	-0.089
only semantic	0.346	-0.094	0.376	-0.097
only skills	0.405	-0.034	0.417	-0.056
only experience	0.417	-0.022	0.461	-0.012
no redaction	0.441	+0.002	0.474	+0.000
no induced vocab	0.435	-0.004	0.468	-0.006
bge encoder	0.426	-0.013	0.483	+0.010

as consistent rather than conclusive. The resume-identity prior and the S1/S2 behaviour of B6 are the stronger evidence.

D. How much labelled data is needed?

Fig. 5 trains on random subsets of training postings. With about 270 pairs M3 is already at 0.396 macro-F1, and it improves to 0.475 with about 5,100 pairs, whereas threshold baselines move from 0.384 to 0.406 and the embedding head from 0.374 to 0.409. Variation across runs is large (standard deviation 0.02–0.06), so differences below about 0.03 at a single size should not be over-read.

E. Are the cited sentences faithful to the model?

For 580 sampled test pairs we delete the sentences the explanation cites (the best sentences of the five strongest met requirements plus the evidence of up to five skills, on average 5.1 sentences) and re-score. The score falls by 7.7 points on average. Deleting the same number of words from randomly chosen other sentences lowers it by 1.8 points. The difference of 5.9 points (95% CI 5.0–6.9) holds for 86.7% of pairs. Cited evidence is therefore not decorative. The test is a check against the model, not a human study of whether reviewers find the explanations useful, which remains future work.

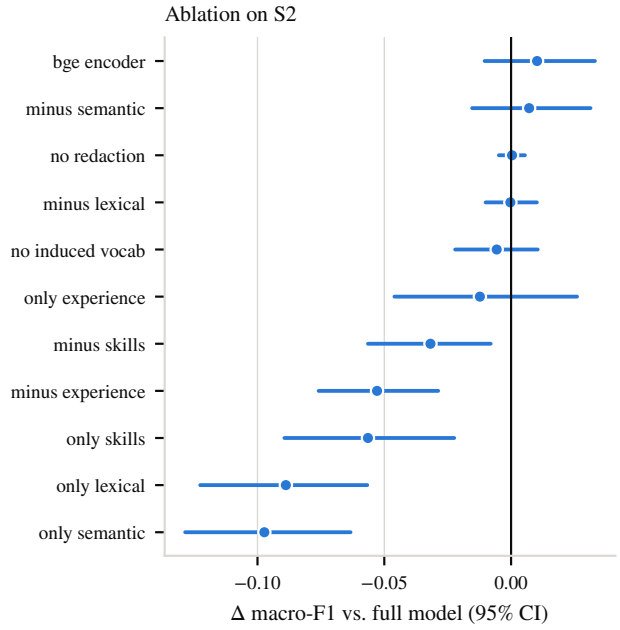


Fig. 4. Change in macro-F1 on S2 when a component is removed or used alone.

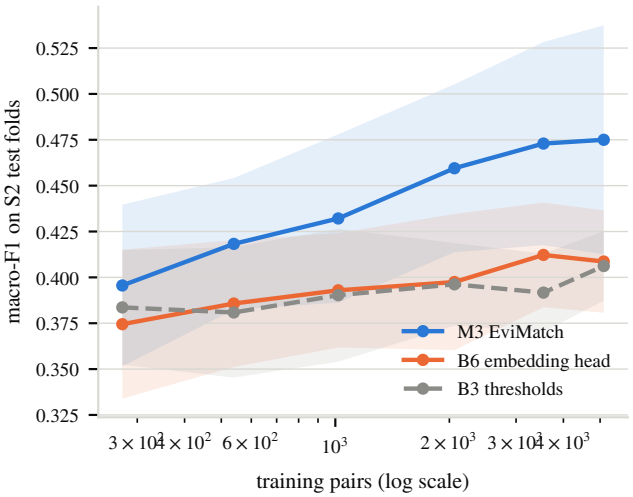


Fig. 5. Macro-F1 on S2 test folds as a function of the number of training pairs (mean of 15 runs, band ±1 s.d.).

F. Counterfactual bias audit

We scored 600 randomly chosen test pairs (69 postings) under 40 names and seven proxy cues (Section III-F), once with the redaction front end and once with it disabled, each time using a linear head trained under the same setting. Table V and Fig. 6 summarise the result; the standard deviation of the score over these pairs is 14.8 points.

TABLE V
Counterfactual audit of M3 on 600 test pairs. Δ score is in points of the 0–100 fit score relative to a neutral control line; flips is the share of pairs whose decision band changes.

	Redaction on		Redaction off	
Cue vs. control line	Δ score [95% CI]	flips	Δ score [95% CI]	flips
affinity org: black	−0.13 [−0.19, −0.08]	1.2%	−0.13 [−0.19, −0.07]	1.3%
affinity org: hispanic	−0.03 [−0.09, +0.03]	1.3%	−0.04 [−0.10, +0.02]	1.0%
affinity org: east asian	−0.22 [−0.33, −0.14]	1.3%	−0.22 [−0.32, −0.13]	1.2%
hobbies: female coded	−0.11 [−0.20, −0.03]	1.2%	−0.12 [−0.20, −0.03]	0.8%
hobbies: male coded	−0.05 [−0.11, +0.00]	1.0%	−0.05 [−0.12, −0.00]	1.0%
career break: childcare	−0.10 [−0.17, −0.05]	1.0%	−0.10 [−0.17, −0.05]	1.0%
named gendered org: female	−0.47 [−0.63, −0.32]	2.2%	−0.09 [−0.18, −0.01]	1.3%
largest gap between name-cell means	0.000		0.154	
largest per-pair change across 40 names	0.000		2.205	

Explicit name cues. With redaction on, the 40 names produce byte-identical redacted text, and the score is identical to the last digit for every one of the 600 pairs. This is true by construction, and it is what the first version of our audit failed to show: the audit had revealed that the header-name pattern could span a newline and then be rejected by a section-header filter, leaving names in place for up to 1.46 points of variation. We fixed the pattern, added regression tests, and re-ran every experiment in this paper; accuracy moved by at most 0.006 macro-F1. With redaction off, the name still has an effect on individual pairs (up to 2.2 points) but not a systematic one: the largest gap between the ten cell means is 0.15 points, and white-male versus black-male and versus black-female differences are not detectable (paired permutation $p = 0.50$ and $p = 0.79$). The absence of a detectable name effect in this model should not be taken as evidence that name-sensitivity is absent in other systems. Our head weights relational features and sees names only as rare tokens, whereas an embedding retriever that ranks by similarity of the whole document lets the name enter the score directly, which is the setting in which large effects were reported [8].

Proxy cues. Redaction does not remove, and does not claim to remove, affinity organisations, hobbies or career-break reasons. Each of these changes the score by a small but non-zero amount relative to a neutral line of similar length: −0.13 points for a Black-affinity organisation, −0.22 for an East-Asian-affinity one, −0.11 for female-coded versus −0.05 for male-coded hobbies (difference $p < 0.001$) and −0.10 for a childcare career break. About 1% of decisions flip. These effects are tiny compared with the score's spread (below 0.02 standard deviations), but they are measurable, are identical with and without redaction, and point in the direction that audit studies of human and machine screeners would lead one to worry about. The row "named gendered org" needs care. With redaction on, the cue is replaced by a placeholder, which differs from the control line, so −0.47 measures the placeholder and not gender; with redaction off the same cue moves the score by −0.09. A cleaner control that passes through the same redactor is a straightforward fix we have not yet run.

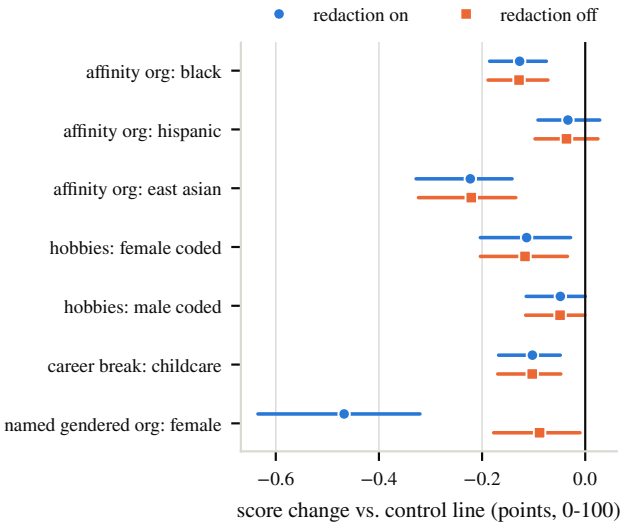


Fig. 6. Score change for each proxy cue relative to its control line, with and without redaction (95% intervals, bootstrap over postings).

VI. Conclusion, Limitations and Future Work

EviMatch shows that a transparent pipeline, trained on 8,000 public pairs, can order candidates better than similarity baselines when resumes and postings are genuinely unseen, while giving a reviewer the resume sentence behind every requirement and the signals behind every score. It also shows what such a system cannot yet do: accuracy is modest, the semantic evidence does not improve it, and redaction leaves measurable proxy effects. The leakage analysis is of independent use, because comparisons on the official split of this benchmark reward a model for recognising resumes.

Limitations. (i) One benchmark, with undocumented label provenance, noisy labels (job-description identity explains 27% of label variance and six of seven duplicated pairs disagree), partly anonymised resumes, and no stated licence; conclusions may not transfer to other labelling schemes. (ii) Absolute accuracy is low enough that outcomes must not be automated. (iii) The skill lexicon is hand-written (350 entries) and English-only, and coverage outside technology and business roles is thin. (iv) Redaction is rule based and removes explicit identifiers; name-to-group assignment in the audit is statistical. (v) We could not evaluate an LLM-only baseline or audit one: the API key available to us had no request quota. Statements about language-model bias therefore rest on the cited literature, not on our measurements. (vi) The deletion test measures faithfulness to the model, not usefulness to people; no user study was run. (vii) The artefact deployed in the prototype is trained on all 8,000 pairs; all reported numbers come from held-out evaluations.

Future work. Concrete next steps are: (a) run the prepared LLM baseline and audit with a pinned open-

weight model and fixed prompts, so that the comparison is reproducible; (b) a study in which recruiters rate whether the displayed evidence changes their decisions and time; (c) validation on a second corpus whose labels come from a documented process; (d) link skills to an open ontology and report extraction coverage by occupation; (e) a pool-normalised ranking mode that uses the other candidates of a posting; and (f) an intersectional audit with larger samples and name lists validated against external data.

Ethics and AI-Assistance Statement

The system is decision support for human reviewers. The benchmark contains resumes of unknown consent status; no personal data beyond the public benchmark were collected, and the prototype does not store uploads. Large-language-model tools were used to help write code and prose for this manuscript; all experiments were run, and every number checked, from the stored result files, and the authors are responsible for the content.

Data and Code Availability

Code, experiment scripts, result files with provenance, the evaluation protocol with its deviation log, and this manuscript's sources are in the project repository (URL to be added at submission). The benchmark is available from its Hugging Face page; it is not redistributed.

References

[1] European Parliament and Council of the European Union, "Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)," Official Journal of the European Union, L 2024/1689, jul 2024.

[2] C. Zhu, H. Zhu, H. Xiong, C. Ma, F. Xie, P. Ding, and P. Li, "Person-Job Fit: Adapting the Right Talent for the Right Job with Joint Representation Learning," ACM Trans. Management Information Systems, 2018.

[3] C. Qin, H. Zhu, T. Xu, C. Zhu, L. Jiang, E. Chen, and H. Xiong, "Enhancing Person-Job Fit for Talent Recruitment: An Ability-aware Neural Network Approach," in Proc. 41st Int. ACM SIGIR Conf. Research and Development in Information Retrieval, 2018.

[4] D. Lavi, V. Medentsiy, and D. Graus, "conSultantBERT: Fine-tuned Siamese Sentence-BERT for Matching Jobs and Job Seekers," arXiv preprint arXiv:2109.06501, 2021.

[5] X. Yu, J. Zhang, and Z. Yu, "ConFit: Improving Resume-Job Matching using Data Augmentation and Contrastive Learning," in Proc. 18th ACM Conf. Recommender Systems (RecSys), 2024, pp. 601–611.

[6] X. Yu, R. Xu, C. Xue, J. Zhang, X. Ma, and Z. Yu, "ConFit v2: Improving Resume-Job Matching using Hypothetical Resume Embedding and Runner-Up Hard-Negative Mining," in Findings of the Association for Computational Linguistics: ACL 2025, 2025, pp. 12 775–12 790.

[7] F. P. W. Lo, J. Qiu, Z. Wang, H. Yu, Y. Chen, G. Zhang, and B. Lo, "AI Hiring with LLMs: A Context-Aware and Explainable Multi-Agent Framework for Resume Screening," arXiv preprint arXiv:2504.02870, 2025.

[8] K. Wilson and A. Caliskan, "Gender, Race, and Intersectional Bias in Resume Screening via Language Model Retrieval," in Proc. AAAI/ACM Conf. AI, Ethics, and Society (AIES), vol. 7, 2024, pp. 1578–1590.

[9] L. Armstrong, A. Liu, S. MacNeil, and D. Metaxa, "The Silicon Ceiling: Auditing GPT's Race and Gender Biases in Hiring," in Proc. 4th ACM Conf. Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO), 2024, pp. 1–18.

[10] H. An, C. Acquaye, C. Wang, Z. Li, and R. Rudinger, "Do Large Language Models Discriminate in Hiring Decisions on the Basis of Race, Ethnicity, and Gender?" in Proc. 62nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), 2024, pp. 386–397.

[11] Z. Wang, Z. Wu, X. Guan, M. Thaler, A. Koshiyama, S. Lu, S. Beepath, E. Ertekin, and M. Perez-Ortiz, "JobFair: A Framework for Benchmarking Gender Hiring Bias in Large Language Models," in Findings of the Association for Computational Linguistics: EMNLP 2024, 2024, pp. 3227–3246.

[12] M. De-Arteaga, A. Romanov, H. Wallach, J. Chayes, C. Borgs, A. Chouldechova, S. Geyik, K. Kenthapadi, and A. T. Kalai, "Bias in Bios: A Case Study of Semantic Representation Bias in a High-Stakes Setting," in Proc. Conf. Fairness, Accountability, and Transparency (FAT*), 2019, pp. 120–128.

[13] H. Gonen and Y. Goldberg, "Lipstick on a Pig: Debiasing Methods Cover up Systematic Gender Biases in Word Embeddings But do not Remove Them," in Proc. 2019 Conf. North American Chapter of the Association for Computational Linguistics (NAACL), 2019, pp. 609–614.

[14] V. Bhatia, P. Rawat, A. Kumar, and R. R. Shah, "End-to-End Resume Parsing and Finding Candidates for a Job Description using BERT," arXiv preprint arXiv:1910.03089, 2019.

[15] K. Khelkhal and D. Lanasri, "Smart-Hiring: An Explainable End-to-End Pipeline for CV Information Extraction and Job Matching," arXiv preprint arXiv:2511.02537, 2025.

[16] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in Proc. 2019 Conf. Empirical Methods in Natural Language Processing and 9th Int. Joint Conf. Natural Language Processing (EMNLP-IJCNLP), 2019, pp. 3980–3990.

[17] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers," in Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020, arXiv:2002.10957.

[18] S. Xiao, Z. Liu, P. Zhang, N. Muennighoff, D. Lian, and J.-Y. Nie, "C-Pack: Packed Resources for General Chinese Embeddings," in Proc. 47th Int. ACM SIGIR Conf. Research and Development in Information Retrieval, 2024, pp. 641–649.

[19] S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," Foundations and Trends in Information Retrieval, vol. 3, no. 4, pp. 333–389, 2009.

[20] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, and I. Gurevych, "BEIR: A Heterogenous Benchmark for Zero-shot Evaluation of Information Retrieval Models," in Proc. NeurIPS Datasets and Benchmarks Track, 2021, arXiv:2104.08663.

[21] O. Khattab and M. Zaharia, "ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT," in Proc. 43rd Int. ACM SIGIR Conf. Research and Development in Information Retrieval, 2020, pp. 39–48.

[22] M. Zhang, K. N. Jensen, S. D. Sonniks, and B. Plank, "SkillSpan: Hard and Soft Skill Extraction from English Job Postings," in Proc. 2022 Conf. North American Chapter of the Association for Computational Linguistics (NAACL), 2022, pp. 4962–4984.

[23] M. Bertrand and S. Mullainathan, "Are Emily and Greg More Employable than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination," American Economic Review, vol. 94, no. 4, pp. 991–1013, 2004.

[24] M. Raghavan, S. Barocas, J. Kleinberg, and K. Levy, "Mitigating Bias in Algorithmic Hiring: Evaluating Claims and Practices," in Proc. 2020 Conf. Fairness, Accountability, and Transparency (FAT*), 2020, pp. 469–481.

[25] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.